



Lab Sheet 1

Database Management System Lab (BTCS203P)

B. TECH AI & ML, CYBERSECURITY, Data Science 3RD Semester (Session 2026-2025)

Objective:

To understand and practice Data Definition Language (DDL) commands for creating and modifying the structure of a table in a relational database using SQL (MySQL).

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

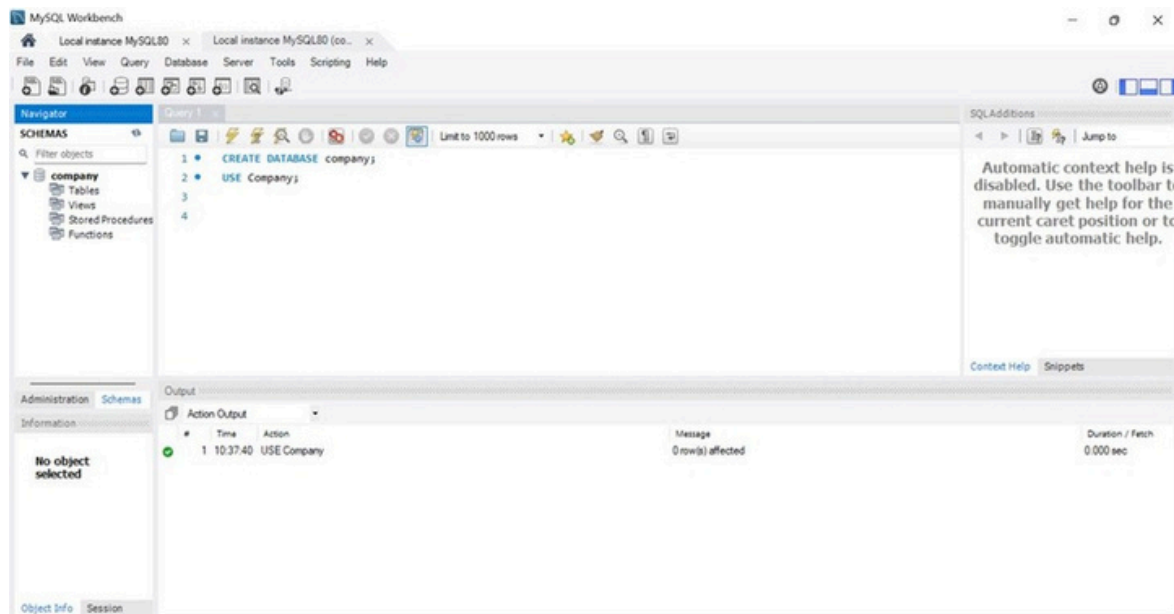
Task: Open SQL client and perform the following tasks.

Task 1: Create a Table

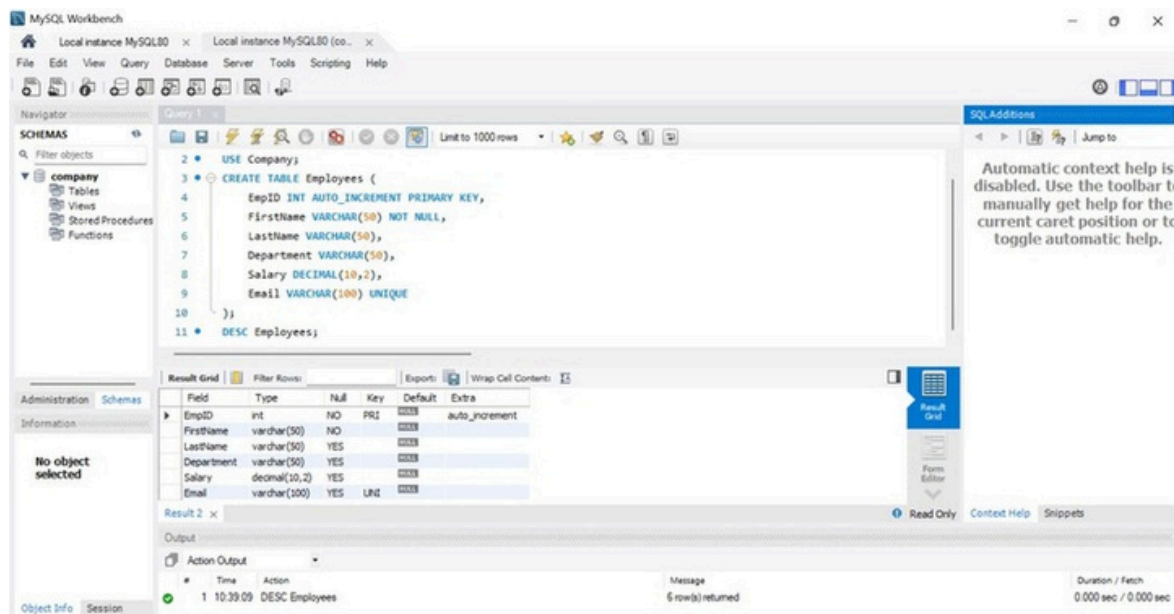
- a. Create a database named Company.
- b. Create a table named Employees with the following structure:

Column Name	Data Type	Constraints
EmpID AUTO_INCREMENT	INT	PRIMARY KEY
FirstName	VARCHAR(50)	NOT NULL
LastName	VARCHAR(50)	
Department	VARCHAR(50)	
Salary	DECIMAL(10,2)	
Email	VARCHAR(100)	UNIQUE

a. Create a database named Company.



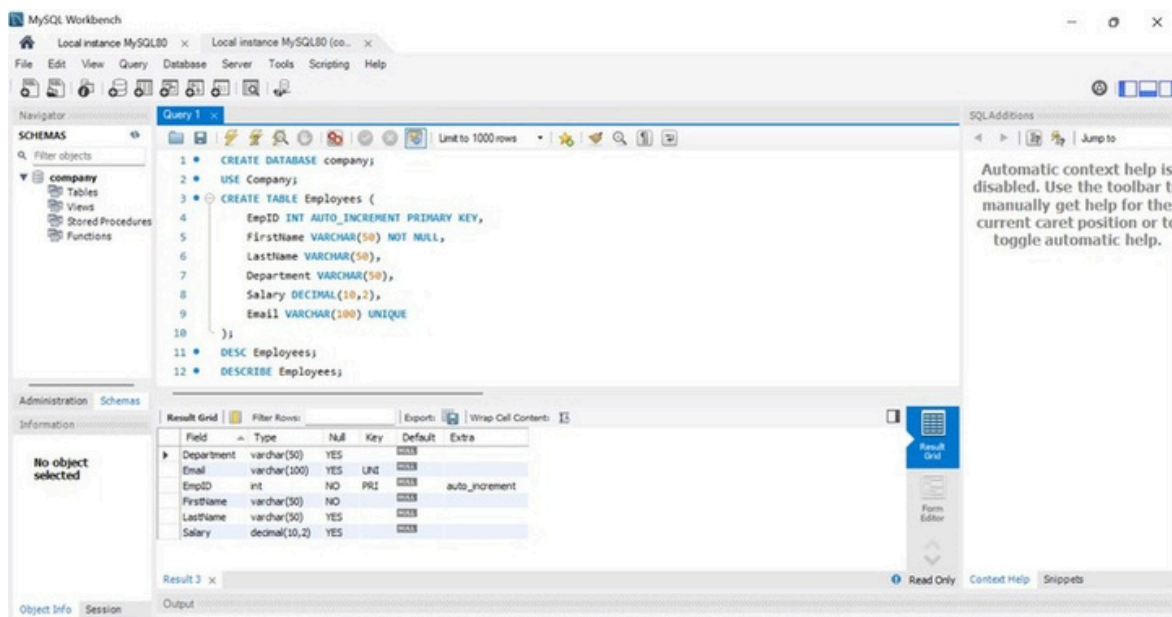
b. Create a table named Employees with the following structure:



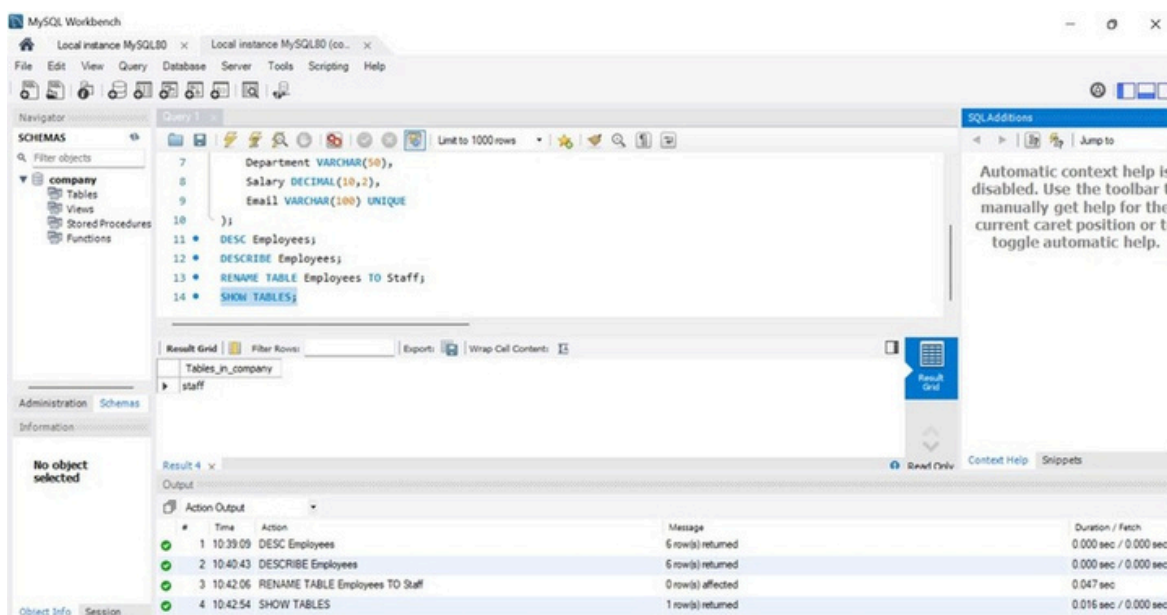
Task 2: DDL Queries on the Employees Table

- a) Describe the structure of the table.
- b) Rename the table to Staff.
- c) Add a new column Phone Number (VARCHAR(15)).
- d) Modify the data type of Salary to DECIMAL(12,2).
- e) Add a DEFAULT value of 'General' to the Department column.
- f) Drop the Phone Number column.
- g) Add a CHECK constraint to ensure Salary is greater than 0.
- h) Rename the column First Name to Given Name.
- i) Drop the chk_Salary constraint.
- j) Create an index on the Email column.
- k) Drop the index on the Email column.
- l) Truncate the table to remove all rows without affecting the structure.
- k) Drop the Staff table completely.

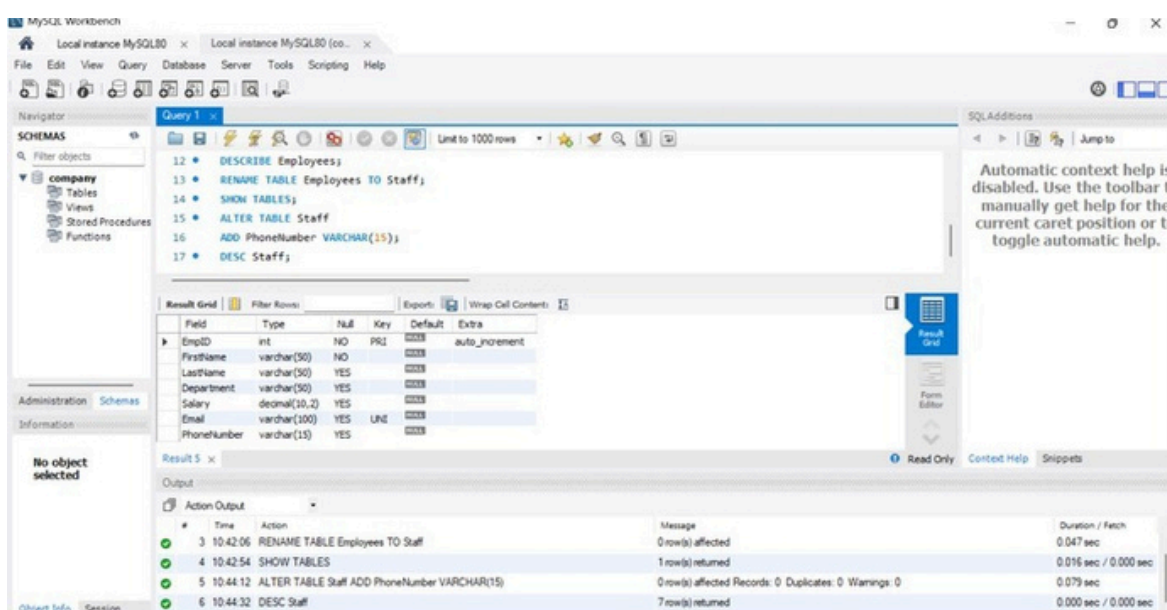
a. Describe the structure of the table.



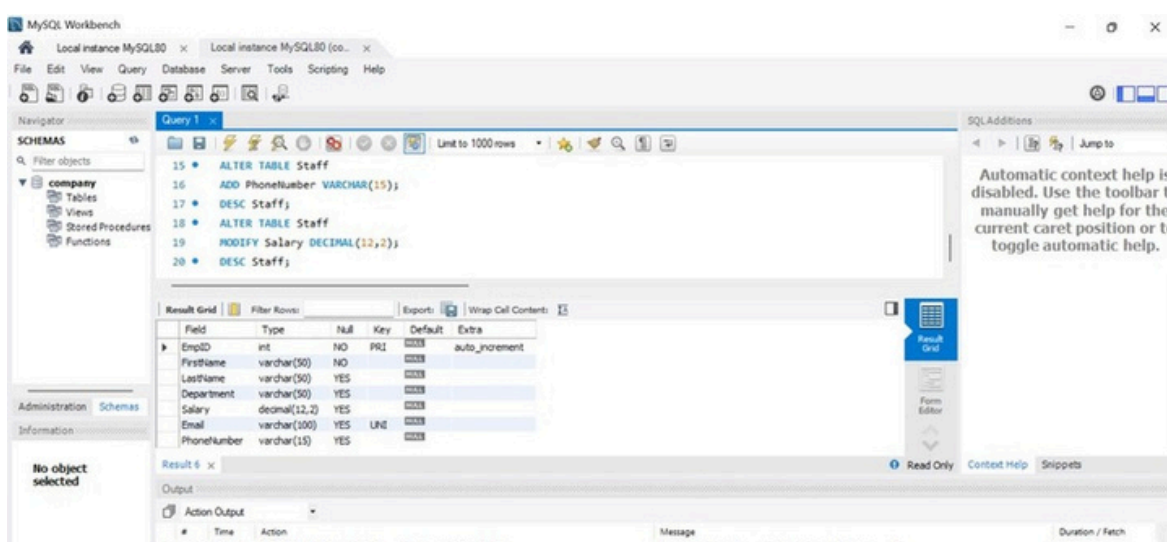
b . Rename the table to Staff.



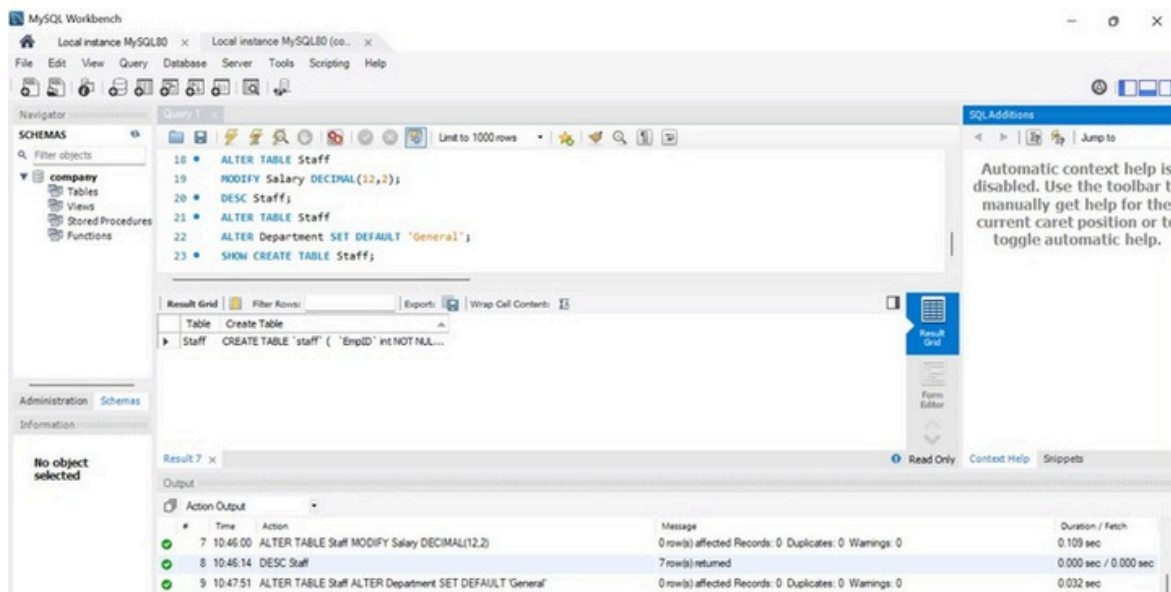
c . AddnewcolumnPhoneNumber(VARCHAR(15)).



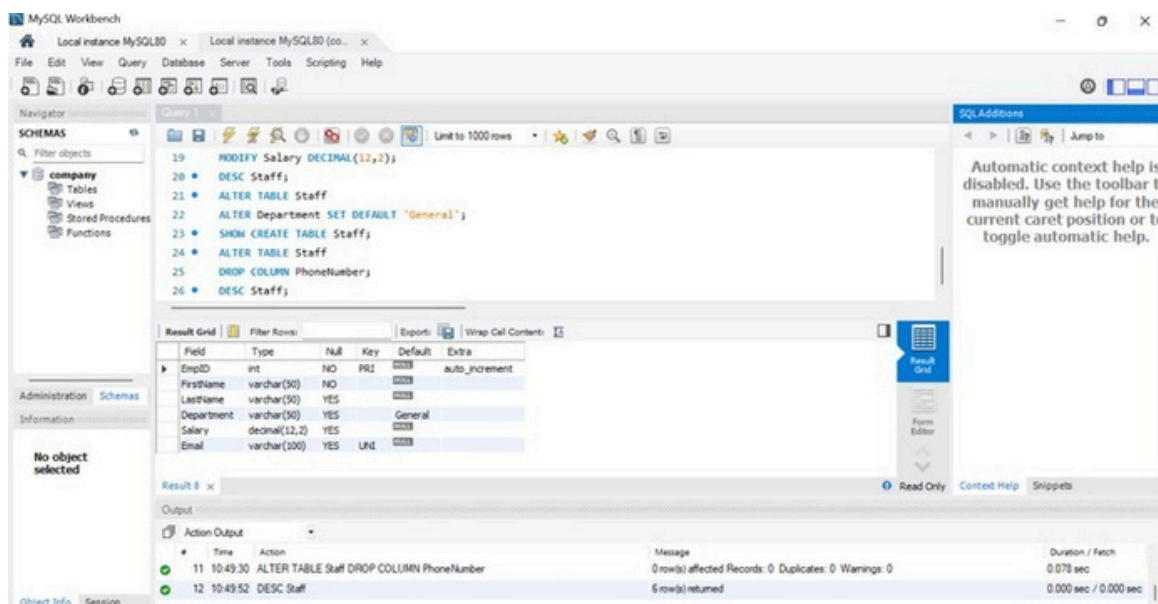
d . Modifythedata typeofSalarytoDECIMAL(12,2).



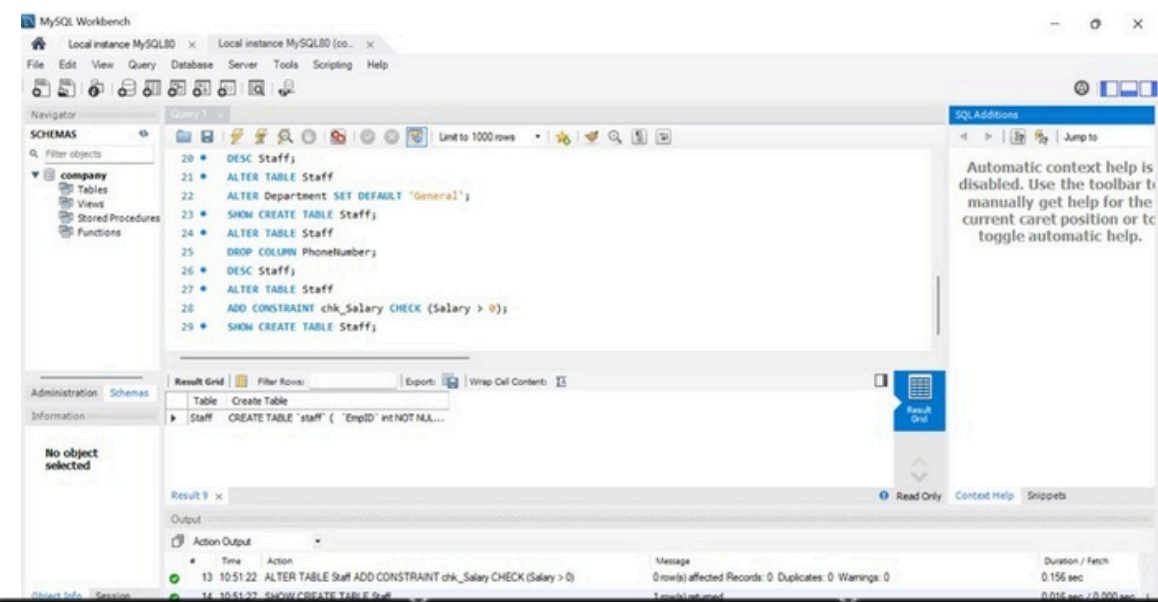
e . AddaDEFAULTvalueof'General'tothe Department column.



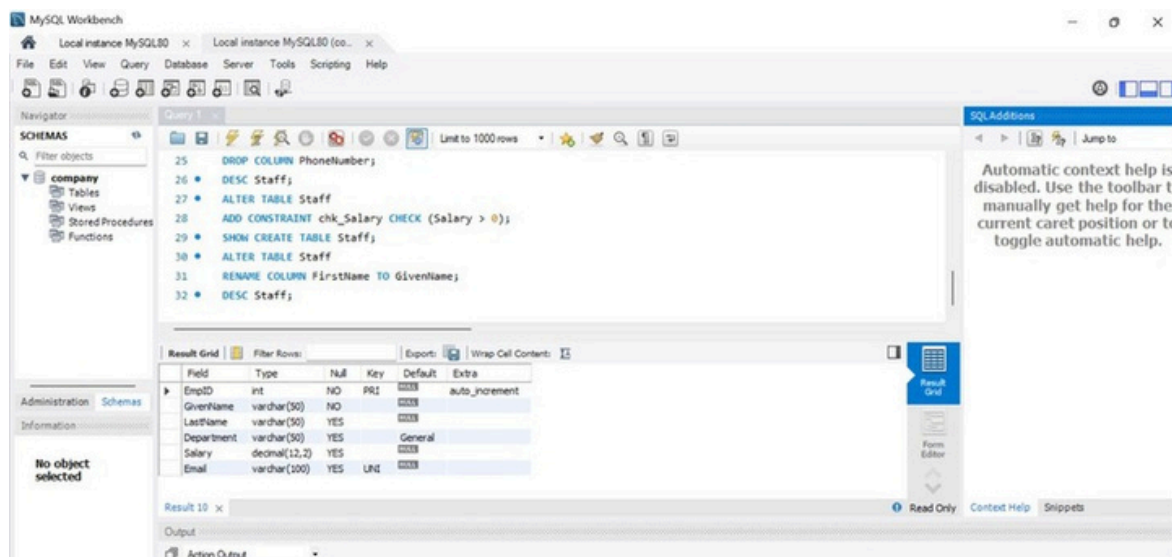
f . Drop the PhoneNumber column.



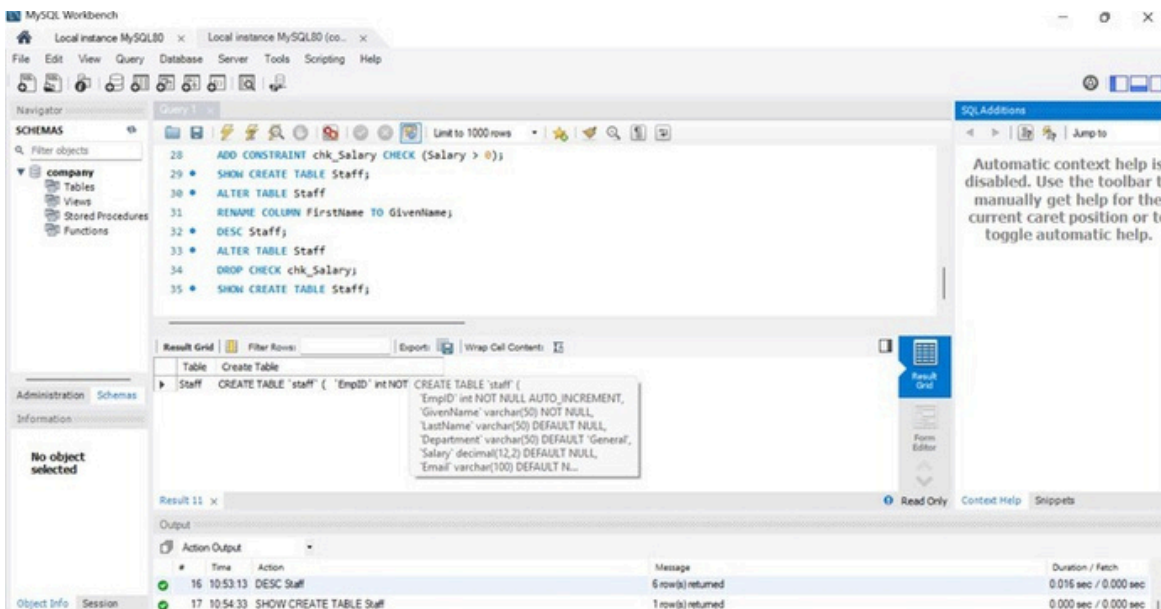
g . AddaCHECKconstrainttoensureSalary is greater than 0.



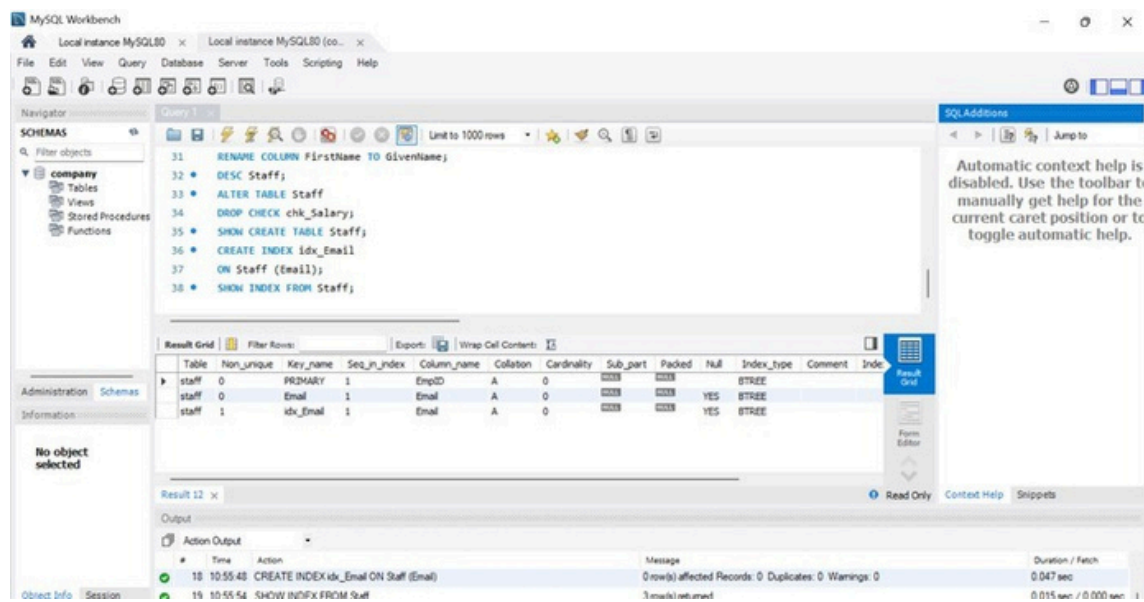
h . .Rename the column FirstName to GivenName.



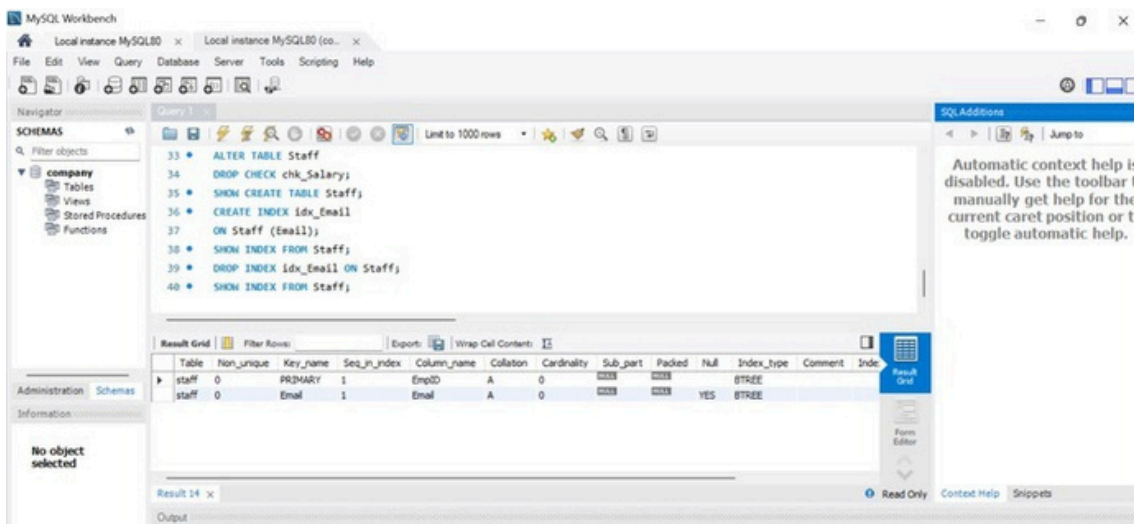
i . Drop the chk_Salary constraint



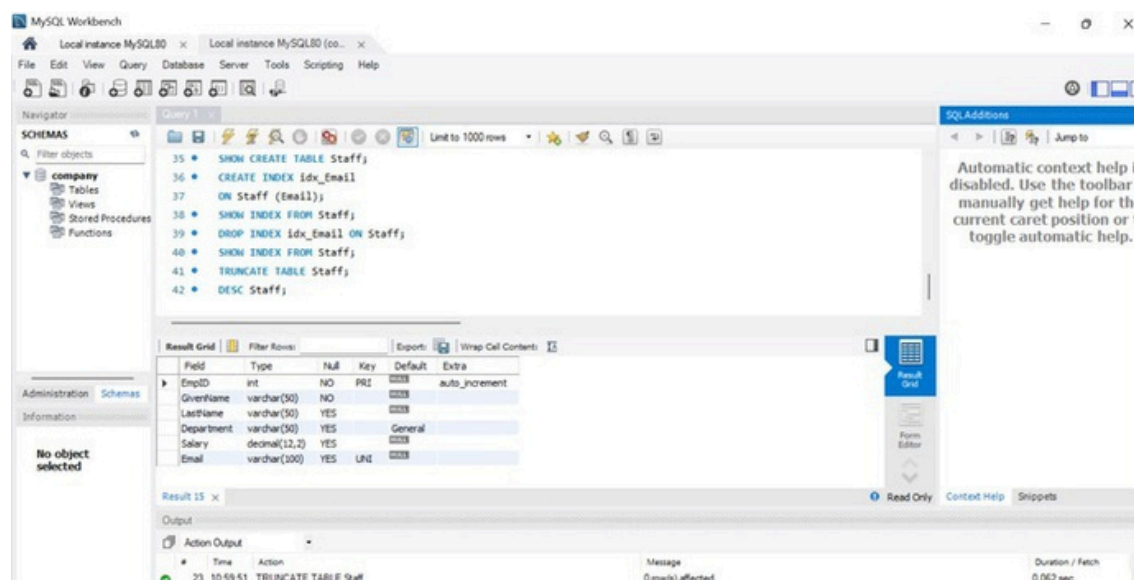
j . Create an index on the Email column.



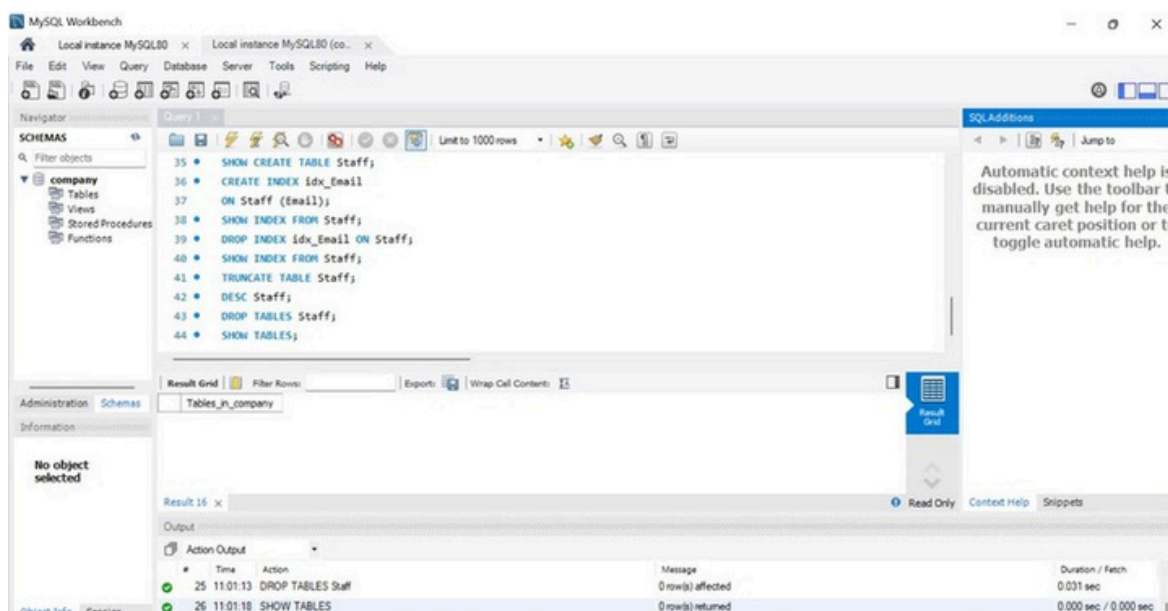
k. . Drop the index on the Email column.



L . Truncate the table to remove all rows without affecting the structure.



M . Drop the Staff table completely.



3. Conceptual Questions:

1. What is the difference between TRUNCATE and DELETE?

DELETE	TRUNCATE
Deletes selected rows or all rows from a table.	Deletes all rows from a table.
Can use WHERE condition.	Cannot use WHERE.
Example: DELETE FROM Student WHERE id=5;	Example: TRUNCATE TABLE Student;
Slower for deleting all records because rows are deleted individually.	Faster because it removes all records at once.
Can be rolled back in many DBMSs when used within a transaction.	Rollback behavior depends on the DBMS.
Table structure remains.	Table structure remains.
DELETE is DML command.	TRUNCATE is generally classified as DDL .

2. Explain the use of the CHECK constraint.

ANS- The CHECK constraint is used to ensure that the values entered into a table satisfy a specific condition.

It prevents invalid data from being inserted into the table.

Syntax:

CREATE TABLE Student (

```
id INT,  
    age INT CHECK (age >=  
    18)  
);
```

Here, CHECK (age >= 18) means that the student's age must be **18 or above**.

Why is it important to define a PRIMARY KEY for a table?

1
ANS- A PRIMARY KEY is important because it uniquely identifies each record (row) in a table.

Main points:

It does not allow duplicate values.

It does not allow NULL values.

It helps identify each record uniquely.

It maintains data integrity in the database.

1. How does ALTER TABLE help in modifying an existing table?

ANS ALTER TABLE is used to modify the structure of an existing table without deleting the table

It can be used to:

1. Add a new column

ALTER TABLE Student ADD age INT;

1. Drop (remove) a column

ALTER TABLE Student
DROP COLUMN age;

1. Modify a column

ALTER TABLE Student MODIFY name
VARCHAR(100);

1. What is the purpose of creating an index in a table?

ANS- An INDEX is created to make data retrieval/searching faster in a table.

For example, if a table has thousands of student records, an index on student_id helps the database find a particular student more quickly.

Example:

```
CREATE INDEX idx_student_name ON Student(name);
```

This creates an index on the name column.

4. Exercises:**1. Create a table named Projects with the following structure:**

Column Name	Data Type	Constraints
ProjectID	INT	PRIMARYKEYAUTO_INCREMENT
NOT NULL		ProjectName VARCHAR(100)
DATE DATE		StartDate DECIMAL(15,2)
CHECK (Budget > 0)		EndDate Budget

- a) Perform at least 5 DDL operations on the Projects table.
- b) Write a query to retrieve the names of all columns in the Staff table.
- c) Write a query to modify the Department column to allow NULL values.

4. Exercises:

1. Create a table named Projects with the following structure:

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Projects' table structure. The right pane shows the 'Output' window with the following message:

#	Time	Action	Message
29	22:53:46	USE company	0 row(s) affected
30	22:54:33	CREATE TABLE Projects (ProjectID INT PRIMARY KEY AUTO_INCREMENT, ProjectName VARCHAR(100) NOT NULL, StartDate DATE, EndDate DATE, Budget DECIMAL(15,2) CHECK (Budget > 0))	0 row(s) affected

2. Perform at least 5 DDL operations on the Projects table.

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Projects' table structure. The right pane shows the 'Output' window with the following messages:

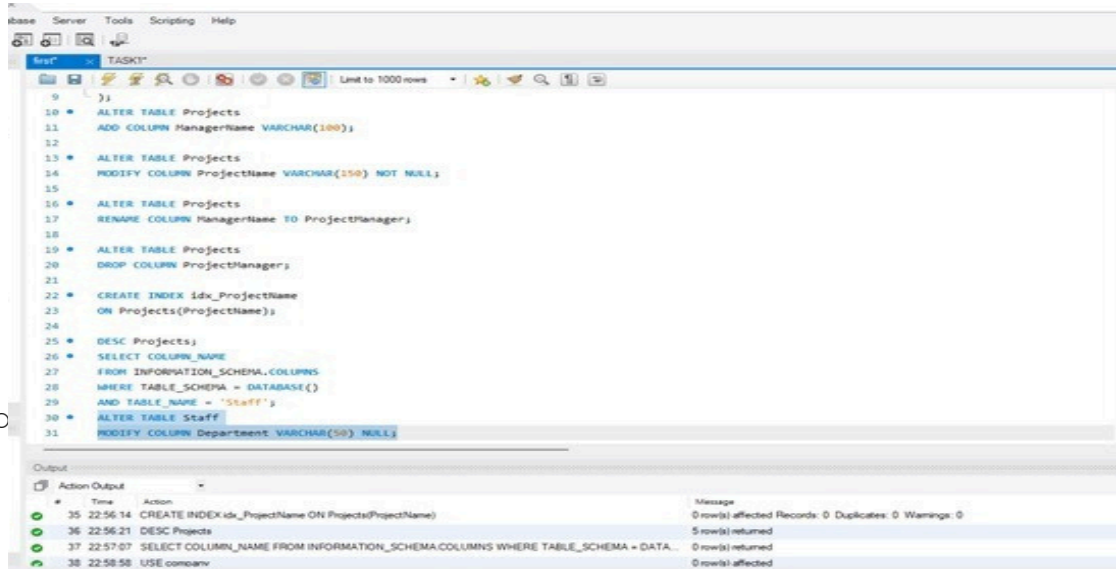
#	Time	Action	Message	Duration / Fetch
32	22:55:56	ALTER TABLE Projects MODIFY COLUMN ProjectName VARCHAR(150) NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
33	22:56:04	ALTER TABLE Projects RENAME COLUMN ManagerName TO ProjectManager	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.032 sec
34	22:56:09	ALTER TABLE Projects DROP COLUMN ProjectManager	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
35	22:56:14	CREATE INDEX idx_ProjectName ON Projects(ProjectName)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

3. Write a query to retrieve the names of all columns in the Staff table.

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Staff' table structure. The right pane shows the 'Output' window with the following messages:

#	Time	Action	Message	Duration / Fetch
38	22:58:58	USE company	0 row(s) affected	0.000 sec
39	22:59:02	CREATE TABLE Projects (ProjectID INT PRIMARY KEY AUTO_INCREMENT, ProjectName VARCHAR(100) NOT NULL, StartDate DATE, EndDate DATE, Budget DECIMAL(15,2) CHECK (Budget > 0))	Error Code: 1050. Table 'projects' already exists	0.000 sec
40	23:06:03	SELECT COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_SCHEMA = 'DATA' AND TABLE_NAME = 'Staff'	0 row(s) returned	0.015 sec / 0.000 sec

4. Write a query to modify the Department column to allow NULL values



The screenshot shows the SQL Server Enterprise Manager interface. The 'TASKS' pane on the left contains a script with the following SQL statements:

```
9  }
10 ALTER TABLE Projects
11 ADD COLUMN ManagerName VARCHAR(100);
12
13 ALTER TABLE Projects
14 MODIFY COLUMN ProjectName VARCHAR(150) NOT NULL;
15
16 ALTER TABLE Projects
17 RENAME COLUMN ManagerName TO ProjectManager;
18
19 ALTER TABLE Projects
20 DROP COLUMN ProjectManager;
21
22 CREATE INDEX idx_ProjectName
23 ON Projects(ProjectName);
24
25 DESC Projects;
26
27 SELECT COLUMN_NAME
28 FROM INFORMATION_SCHEMA.COLUMNS
29 WHERE TABLE_SCHEMA = DATABASE()
30 AND TABLE_NAME = 'Staff';
31
32 ALTER TABLE Staff
33 MODIFY COLUMN Department VARCHAR(50) NULL;
```

The 'Output' pane at the bottom shows the execution results of the script:

#	Time	Action	Message
35	22:56:14	CREATE INDEX idx_ProjectName ON Projects(ProjectName)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
36	22:56:21	DESC Projects	5 row(s) returned
37	22:57:07	SELECT COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_SCHEMA = DATA...	0 row(s) returned
38	22:58:58	USE company	0 row(s) affected